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Patent Public Search | Text View

United States Patent Application Publication

20250285268

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

KAMON; Shumpei

MEDICAL IMAGE PROCESSING DEVICE, HIERARCHICAL NEURAL NETWORK, MEDICAL IMAGE PROCESSING METHOD, AND PROGRAM

Abstract

There are provided a medical image processing device, a hierarchical neural network, a medical image processing method, and a program capable of achieving highly accurate and real-time processable region-of-interest detection and class classification. A medical image processing device acquires a medical image, extracts a first feature amount and a second feature amount having a resolution relatively higher than a resolution of the first feature amount from the medical image by processing the medical image in a feature extraction network of a hierarchical neural network, detects a region of interest included in the medical image by processing the first feature amount in a first subnetwork of the hierarchical neural network, and classifies the region of interest by processing the second feature amount in a second subnetwork of the hierarchical neural network.

Inventors: KAMON; Shumpei (Kanagawa, JP)

Applicant: FUJIFILM Corporation (Tokyo, JP)

Family ID: 96898299

Assignee: FUJIFILM Corporation (Tokyo, JP)

Appl. No.: 19/061950

Filed: February 24, 2025

Foreign Application Priority Data

JP 2024-033022

Mar. 05, 2024

Publication Classification

Int. Cl.: G06T7/00 (20170101); **G06T7/11** (20170101); **G06V10/25** (20220101); **G06V10/82** (20220101)

U.S. Cl.:

CPC G06T7/0012 (20130101); **G06T7/11** (20170101); **G06V10/25** (20220101); **G06V10/82** (20220101); **G06T2207/20132** (20130101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] The present application claims priority under 35 U.S.C § 119(a) to Japanese Patent Application No. 2024-033022 filed on Mar. 5, 2024, which is hereby expressly incorporated by reference, in its entirety, into the present application.

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0002] The present invention relates to a medical image processing device, a hierarchical neural network, a medical image processing method, and a program, and particularly relates to a technique of detecting and classifying a region of interest from a medical image.

2. Description of the Related Art

[0003] A function of detecting a region of interest such as a lesion from a medical image captured by a medical image diagnostic apparatus such as an endoscope and an ultrasound diagnostic apparatus and simultaneously classifying the region of interest into a class such as a disease type is known. Such a function can be implemented by training a neural network (NN) using an image including the region of interest and position information and classification class information of the region of interest (see WO2020/203552A and WO2019/142243A).

[0004] During inference, feature extraction is performed while gradually reducing a resolution of an input image using NN processing, and the region detection and classification are performed from the obtained feature amount. In the classification task, the classification processing is performed after cropping a part corresponding to the detection region from the feature amount reduced in resolution. Therefore, a processing target in the classification task has a small feature resolution, which leads to a decrease in accuracy in a case in which it is necessary to evaluate a detailed structure such as lesion classification. In particular, in a model designed for real-time processing, this problem is conspicuous because it is necessary to actively reduce the resolution for high-speed processing.

SUMMARY OF THE INVENTION

[0005] As a method of solving such a problem, a method of cropping a detection region from an input image and training a classification task using the cropped image is considered. However, it is inefficient to perform the feature extraction using the NN processing again from a high-resolution input image, which may make the real-time processing difficult.

[0006] The present invention has been made in view of such circumstances, and an object of the present invention is to provide a medical image processing device, a hierarchical neural network, a medical image processing method, and a program capable of achieving highly accurate and real-time processable region-of-interest detection and class classification.

[0007] In order to achieve the above object, a first aspect of the present disclosure provides a medical image processing device comprising: one or more processors that acquire a medical image; and one or more memories that store a program to be executed by the one or more processors, in which the one or more processors extract a first feature amount and a second feature amount having

a resolution relatively higher than a resolution of the first feature amount from the medical image by processing the medical image in a feature extraction network of a hierarchical neural network including the feature extraction network, a first subnetwork, and a second subnetwork, detect a region of interest included in the medical image by processing the first feature amount in the first subnetwork of the hierarchical neural network, and classify the region of interest by processing the second feature amount in the second subnetwork of the hierarchical neural network.

[0008] A second aspect of the present disclosure provides the medical image processing device according to the first aspect, in which it is preferable that the second feature amount is an intermediate feature amount in a process of extracting the first feature amount from the medical image in the feature extraction network.

[0009] A third aspect of the present disclosure provides the medical image processing device according to the first or second aspect, in which it is preferable that the one or more processors train the first subnetwork using a first data set and train the second subnetwork using a second data set, the first data set includes a set of a first medical image and position information of a region of interest included in the first medical image, and the second data set includes a set of a second medical image and position information and a classification class label of a region of interest included in the second medical image.

[0010] A fourth aspect of the present disclosure provides the medical image processing device according to the third aspect, in which it is preferable that the one or more processors train the feature extraction network and the first subnetwork using the first data set, and train the second subnetwork using the second data set based on the trained feature extraction network and the trained first subnetwork.

[0011] A fifth aspect of the present disclosure provides the medical image processing device according to the fourth aspect, in which it is preferable that the one or more processors pre-train the feature extraction network using a third data set different from the first data set and the second data set before training the feature extraction network and the first subnetwork using the first data set.

[0012] A sixth aspect of the present disclosure provides the medical image processing device according to any one of the first to fifth aspects, in which it is preferable that the one or more processors notify of position information of the region of interest in a manner corresponding to a result of the classification.

[0013] A seventh aspect of the present disclosure provides the medical image processing device according to the sixth aspect, in which it is preferable that the one or more processors add information based on the position information to the medical image and display the medical image on a display.

[0014] An eighth aspect of the present disclosure provides the medical image processing device according to the sixth or seventh aspect, in which it is preferable that the one or more processors do not notify of the position information of the region of interest in a case in which the result of the classification is a specific class.

[0015] A ninth aspect of the present disclosure provides the medical image processing device according to the eighth aspect, in which it is preferable that classes of the classification include a malignancy grade, and the one or more processors do not notify of the position information of the region of interest in a case in which the malignancy grade is relatively low.

[0016] A tenth aspect of the present disclosure provides the medical image processing device according to any one of the first to ninth aspects, in which it is preferable that the one or more processors crop a part of the second feature amount according to a detection result of the region of interest, and process the cropped second feature amount in the second subnetwork.

[0017] An eleventh aspect of the present disclosure provides the medical image processing device according to the tenth aspect, in which it is preferable that the one or more processors align a size of the cropped second feature amount in a spatial direction to a certain size, and process the second feature amount having the certain size in the second subnetwork.

[0018] In order to achieve the above object, a twelfth aspect of the present disclosure provides a hierarchical neural network comprising: a feature extraction network that extracts a first feature amount and a second feature amount having a resolution relatively higher than a resolution of the first feature amount from an input medical image; a first subnetwork that detects a region of interest included in the medical image from the input first feature amount; and a second subnetwork that classifies the region of interest from the input second feature amount.

[0019] In order to achieve the above object, a thirteenth aspect of the present disclosure provides a medical image processing method executed by one or more processors, the medical image processing method comprising: acquiring a medical image; extracting a first feature amount and a second feature amount having a resolution relatively higher than a resolution of the first feature amount from the medical image by processing the medical image in a feature extraction network of a hierarchical neural network including the feature extraction network, a first subnetwork, and a second subnetwork; detecting a region of interest included in the medical image by processing the first feature amount in the first subnetwork of the hierarchical neural network; and classifying the region of interest by processing the second feature amount in the second subnetwork of the hierarchical neural network.

[0020] In order to achieve the above object, a fourteenth aspect of the present disclosure provides a program for causing a computer to execute the medical image processing method according to the thirteenth aspect. The present disclosure also includes a non-transitory computer-readable storage medium in which the program according to the thirteenth aspect is stored.

[0021] According to the present invention, it is possible to achieve highly accurate and real-time processable region-of-interest detection and class classification.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0022] FIG. 1 is a conceptual diagram of a network system that performs general detection processing.

[0023] FIG. 2 is a conceptual diagram of a network system in which detection and classification are performed using separate NNs.

[0024] FIG. 3 is a conceptual diagram of a network system according to a first embodiment.

[0025] FIG. 4 is a flowchart showing steps of a medical image processing method.

[0026] FIG. 5 is a block diagram showing a configuration of a medical image processing device.

[0027] FIG. 6 is a schematic diagram showing an overall configuration of an endoscope system including the medical image processing device.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0028] Hereinafter, preferred embodiments of the present invention will be described with reference to the accompanying drawings. The same components are denoted by the same reference numerals, and overlapping description will not be repeated.

General Detection Processing

[0029] FIG. 1 is a conceptual diagram of a network system **10** that performs general detection processing. The network system **10** includes a feature extraction unit **12** and a detection processing unit **14**.

[0030] The feature extraction unit **12** is an NN that outputs a feature amount of an image in a case in which the image is input. In the example shown in FIG. 1, the feature extraction unit **12** receives a medical image as an input image IM as an input and outputs a feature amount of the input image IM. In FIG. 1, an arrow represents NN processing, FV1, FV2, FV3, and FV4 represent feature amounts of a processing process, and horizontal widths of FV1, FV2, FV3, and FV4 represent a size of a resolution in a spatial direction. The feature extraction unit **12** extracts features of the

input image IM while gradually decreasing the resolution of the feature amount. The feature amount extracted by the feature extraction unit **12** is input to the detection processing unit **14**. [0031] The detection processing unit **14** is an NN that simultaneously outputs a position of a lesion, which is a region of interest of the image, and class classification of the lesion in a case in which the feature amount of the image is input. In the example shown in FIG. **1**, the detection processing unit **14** receives the feature amount FV**4** extracted by the feature extraction unit **12** as an input, and outputs a bounding box BB indicating the position of the lesion in the input image IM and a result of the class classification of the lesion, "Class: cancer".

[0032] In the network system **10**, it is necessary to perform the class classification of the region of interest based on the feature amount having a relatively low resolution, which leads to a decrease in accuracy in a task such as lesion classification in which a detailed structure of the surface needs to be evaluated.

Detection Processing of Performing Detection and Classification Separately

[0033] As a method of increasing the resolution during classification, there is a method of performing detection and classification in separate NNs. FIG. **2** is a conceptual diagram of a network system **20** in which detection and classification are performed in separate NNs. The network system **20** includes a feature extraction unit **22**, a detection processing unit **24**, a cropping unit **26**, and a classification processing unit **28**.

[0034] The feature extraction unit **22** is an NN that outputs a feature amount of an image in a case in which the image is input. In the example shown in FIG. **2**, the feature extraction unit **22** receives a medical image as an input image IM as an input and outputs a feature amount of the input image IM. In FIG. **2**, a solid arrow represents NN processing, FV**11**, FV**12**, FV**13**, and FV**14** represent feature amounts of a processing process, and horizontal widths of FV**11**, FV**12**, FV**13**, and FV**14** represent a size of a resolution in a spatial direction. The feature extraction unit **22** extracts features of the input image IM while gradually decreasing the resolution of the feature amount. The feature amount extracted by the feature extraction unit **22** is input to the detection processing unit **24**.

[0035] The detection processing unit **24** is an NN that outputs only the position of the lesion of the image in a case in which the feature amount of the image is input. In the example shown in FIG. **2**, the detection processing unit **24** receives the feature amount FV**14** extracted by the feature extraction unit **22** as an input, and outputs a bounding box BB indicating the position of the lesion in the input image IM.

[0036] In a case in which the input image IM and the bounding box BB are input, the cropping unit **26** generates a cropped region CR obtained by cropping a region surrounded by the bounding box BB from the input image IM.

[0037] The classification processing unit **28** is an NN that outputs class classification of a disease type of the lesion included in the cropped region in a case in which the cropped region of the image is input. In the example shown in FIG. **2**, the cropped region CR is input, and a result of the class classification of the lesion, "Class: cancer", is output. In FIG. **2**, a solid arrow represents NN processing, FV**21**, FV**22**, and FV**23** represent feature amounts of a processing process, and horizontal widths of FV**21**, FV**22**, and FV**23** represent a size of a resolution in a spatial direction.

[0038] With the network system **20**, the problem of the resolution in the classification processing is solved. However, since it is necessary to perform the feature extraction again from the input image IM in the classification processing unit **28**, the calculation efficiency is poor, and it is inappropriate in a situation where real-time processing is required.

First Embodiment

Configuration of Network System

[0039] FIG. **3** is a conceptual diagram of a network system **30** according to a first embodiment. The network system **30** includes a feature extraction unit **22**, a detection processing unit **24**, a cropping unit **36**, a resizing unit **38**, and a classification processing unit **40**.

[0040] The feature extraction unit **22** is an NN that extracts a first feature amount and a second

feature amount having a resolution relatively higher than a resolution of the first feature amount from the input medical image. The second feature amount may be an intermediate feature amount in a process of extracting the first feature amount from the input image through the feature extraction unit **22**. The resolution of the second feature amount need only be decided in consideration of a trade-off between the accuracy of the classification of the lesion and the time required for the classification. The feature extraction unit **22** holds an intermediate feature amount in a calculation process while estimating only position information of the lesion, which is a region of interest, through the detection processing. In the example shown in FIG. **3**, FV**14** corresponds to the first feature amount, and FV**12** corresponds to the second feature amount.

[0041] The detection processing unit **24** is an NN that outputs only the position of the lesion of the image in a case in which the feature amount of the image is input. In the example shown in FIG. **3**, the detection processing unit **24** receives the feature amount FV**14** as the first feature amount as an input, and outputs a bounding box BB indicating the position of the lesion in the input image IM.

[0042] The cropping unit **36** crops a part of the second feature amount according to a detection result of the region of interest, and generates a feature map by cropping a region corresponding to the bounding box BB from a feature map indicated by the feature amount in a case in which the feature amount of the image and the bounding box BB are input. In the example shown in FIG. **3**, the cropping unit **36** receives a feature map FM**1** indicated by the feature amount FV**12** which is the second feature amount and the bounding box BB as an input, and outputs a feature map FM**2** obtained by cropping a region corresponding to the bounding box BB from the feature map FM**1**.

[0043] The feature map FM**1** is two-dimensional data having a size of width in a horizontal direction and a size of height in a vertical direction. The feature map FM**1** indicates a feature amount of the input image IM in which the position information in the input image IM is reflected. The feature map FM**2** indicates a feature amount of the lesion in the input image IM.

[0044] The resizing unit **38** generates a feature map FM**3** by aligning a size of the input feature map FM**2** in the spatial direction to a constant size by interpolation processing or the like. The certain size is, for example, a size suitable for input to the classification processing unit **40**.

[0045] The classification processing unit **40** is an NN that classifies the disease type of the lesion from the input feature amount. In the example shown in FIG. **3**, the classification processing unit **40** receives the feature map FM**3** as an input, and outputs a result of the class classification of the lesion included in the feature map FM**3**, "Class: cancer". In FIG. **3**, a solid arrow represents NN processing, FV**31**, FV**32**, and FV**33** represent feature amounts of a processing process, and horizontal widths of FV**31**, FV**32**, and FV**33** represent a size of a resolution in a spatial direction. The classification processing unit **40** may perform class classification of the malignancy grade of the lesion.

[0046] As described above, the network system **30** includes a hierarchical neural network comprising the feature extraction unit **22** which is a feature extraction network, the detection processing unit **24** which is a first subnetwork, and the classification processing unit **40** that is a second subnetwork.

Medical Image Processing Method

[0047] FIG. **4** is a flowchart showing steps of a medical image processing method using the network system **30**.

[0048] In step S**1**, the network system **30** acquires a medical image. In the example shown in FIG. **3**, the input image IM is acquired.

[0049] In step S**2**, the network system **30** extracts a first feature amount and a second feature amount having a resolution relatively higher than a resolution of the first feature amount by processing the medical image acquired in step S**1** in the feature extraction unit **22**. In the example shown in FIG. **3**, FV**14** is extracted as the first feature amount, and FV**12** is extracted as the second feature amount.

[0050] In step S**3**, the network system **30** detects the lesion included in the medical image by

processing the first feature amount extracted in step S2 in the detection processing unit 24. In the example shown in FIG. 3, the bounding box BB indicating the position of the lesion of the input image IM is output.

[0051] In step S4, the network system 30 crops the region of the lesion detected in step S3 from the second feature amount extracted in step S2 by using the cropping unit 36. In the example shown in FIG. 3, the feature map FM2 is cropped.

[0052] In step S5, the network system 30 resizes the feature map cropped in step S4 to a certain size by using the resizing unit 38. In the example shown in FIG. 3, the feature map FM3 is generated.

[0053] In step S6, the network system 30 performs class classification of the lesion detected in step S3 by processing the feature map resized in step S5 in the classification processing unit 40. That is, the network system 30 performs class classification of the lesion by processing the second feature amount. In the example shown in FIG. 3, the result of the class classification of the lesion, "Class: cancer", is output.

[0054] As described above, according to the medical image processing method, the intermediate feature amount of the feature extraction unit 22 is used instead of cropping from the input image IM. This method enables classification processing based on the feature amount having a higher resolution than the network system 10, and does not require the feature extraction using the NN processing again from the input image as in the network system 20. As a result, it is possible to achieve both high accuracy and real-time processing. In addition, the network system 30 performs the classification processing after aligning the sizes of the feature maps in the spatial direction, so that it is possible to reduce the influence of the size difference of the region of interest.

[0055] Here, the second feature amount is the intermediate feature amount in the process of extracting the first feature amount from the input image through the feature extraction unit 22, but the method of extracting the second feature amount is not limited to this example. For example, in the feature extraction unit 22 having decoding processing of increasing a resolution, such as U-net, the resolution may be increased from the first feature amount to generate the second feature amount. In addition, in the feature extraction unit 22 where the NN branches in the middle, the first feature amount and the second feature amount may be extracted separately.

Training of Hierarchical Neural Network

[0056] For training of the hierarchical neural network consisting of the feature extraction unit 22, the detection processing unit 24, and the classification processing unit 40, a plurality of sets of learning data sets in which a training image and correct answer data are combined are used.

[0057] The training image may be, for example, a medical image. The medical image may be, for example, an endoscope image, a computed tomography (CT) image, a magnetic resonance imaging (MRI) image, and an ultrasound image.

[0058] The correct answer data includes position information of a region of interest included in the training image and classification label information of the region of interest.

[0059] The detection processing unit 24 may be trained using a first data set, and the classification processing unit 28 may be trained using a second data set different from the first data set. The first data set includes a set of a first medical image and position information of a region of interest included in the first medical image, and the second data set includes a set of a second medical image and position information and a classification class label of a region of interest included in the second medical image.

[0060] The training of the detection processing unit 24 requires a pair of the training image and the position information of the region of interest, and the training of the classification processing unit 40 further requires information on a classification label of the region of interest. In a case in which the entire hierarchical neural network is trained using the same data set, it is necessary to prepare correct answer data for all training images. On the other hand, in a method of separately training the detection processing unit 24 and the classification processing unit 40 by using different data

sets, it is not necessary to add classification labels to all data. It is sufficient to train only the detection processing unit **24** by using the learning data set to which the position information is added as the correct answer data, and to train the classification processing unit **40** by using only the learning data set to which the classification label is added as the correct answer data.

[0061] In a case of performing learning in two stages as described above, it is desirable to train the feature extraction unit **22** and the detection processing unit **24** first, and then train the classification processing unit **40** in a state where learned weights are transferred to the feature extraction unit **22** and the detection processing unit **24**. That is, the feature extraction unit **22** and the detection processing unit **24** are trained using the first data set, and the classification processing unit **28** is trained using the second data set based on the trained feature extraction unit **22** and the trained detection processing unit **24**.

[0062] Before training the feature extraction unit **22** and the detection processing unit **24**, the feature extraction unit **22** may be pre-trained using a third data set different from the first data set and the second data set. The third data set includes a set of an image and a result of a task of the image. The image is not limited to a medical image and may be a general image. One advantage of the general image is that it is easier to obtain a large amount of data than the medical image. In addition, the task is not limited to the detection task and may be a classification task.

Configuration of Medical Image Processing Device

[0063] FIG. **5** is a block diagram showing a configuration of a medical image processing device **50** to which the network system **30** is applied. The medical image processing device **50** is implemented by at least one computer. As shown in FIG. **5**, the medical image processing device **50** comprises a processor **52**, a memory **54**, an input device **56**, and an output device **58**.

[0064] The processor **52** acquires a medical image and executes a command stored in the memory **54**. A hardware structure of the processor **52** is various processors as shown below. The various processors include a central processing unit (CPU) that is a general-purpose processor acting as various functional units including the cropping unit **36** and the resizing unit **38** by executing software (program), a graphics processing unit (GPU) that is a processor specialized in image processing, a programmable logic device (PLD) that is a processor of which a circuit configuration is changeable after manufacturing, such as a field programmable gate array (FPGA), a dedicated electric circuit that is a processor having a circuit configuration dedicatedly designed to execute a specific process, such as an application specific integrated circuit (ASIC), or the like.

[0065] One processing unit may be configured of one of these various processors, or may be configured of two or more processors of the same type or different types (for example, a plurality of FPGAs, a combination of a CPU and an FPGA, or a combination of a CPU and a GPU). Further, a plurality of functional units may be configured of one processor. As an example in which the plurality of functional units are configured of one processor, first, as typified by a computer such as a client or a server, one processor is configured of a combination of one or more CPUs and software and this processor acts as the plurality of functional units. Second, as typified by a system on chip (SoC) or the like, a processor that realizes the functions of the entire system including the plurality of functional units with one integrated circuit (IC) chip is used. As described above, the various functional units are configured by using one or more of the above described various processors as a hardware structure.

[0066] The hardware structure of these various processors is more specifically an electric circuit (circuitry) in which circuit elements such as semiconductor elements are combined.

[0067] The memory **54** stores a command to be executed by the processor **52**. In addition, the memory **54** stores a program and a weight parameter for causing the NN of each of the feature extraction unit **22**, the detection processing unit **24**, and the classification processing unit **40** to function. The memory **54** includes a random access memory (RAM) and a read only memory (ROM), neither of which is shown. The processor **52** executes various types of processing of the medical image processing device **50** by using the RAM as a work region, executing software by

using various programs and parameters stored in the ROM, and using the parameters stored in the ROM and the like.

[0068] The medical image processing method shown in FIG. 4 is implemented by the processor 52 executing a medical image processing program stored in the memory 54. The medical image processing program may be provided by a computer-readable non-transitory storage medium. In this case, the medical image processing device 50 may read the medical image processing program from the non-transitory storage medium and store the medical image processing program in the memory 54.

[0069] The medical image processing device 50 may train the hierarchical neural network by the processor 52 executing a learning program stored in the memory 54. The learning data set may be stored in the memory 54. The hierarchical neural network may be trained by a computer different from the medical image processing device 50. In this case, a program and a weight parameter for causing the trained NN to function need only be stored in the memory 54.

[0070] The input device 56 is configured of, for example, a keyboard, a mouse, a touch panel, or other pointing devices, or a voice input device, or an appropriate combination thereof. A doctor can input various instructions and information to the medical image processing device 50 using the input device 56.

[0071] The output device 58 includes a display device. For example, the display device may be a liquid crystal display, an organic electro-luminescence (OEL) display, a projector, or an appropriate combination thereof. The input device 56 and the display device of the output device 58 may be integrally configured, such as a touch panel. In addition, the output device 58 may include a voice output device such as a speaker that outputs a voice. The medical image processing device 50 can output the result of the processing of the detection and the class classification of the lesion to the output device 58 and notify the doctor of the result using image information or a voice.

Second Embodiment

[0072] In a case in which the medical image processing device 50 notifies the doctor of the result of the processing of the detection and the class classification of the lesion using the image information or the voice, the doctor may be notified of the result in different aspects according to the classification result. For example, in a case in which the medical image processing device 50 displays a figure indicating the position of the lesion together with the input image on the display device, the medical image processing device 50 makes at least one of a color, a line type, a thickness, the presence or absence of blinking, a blinking frequency, or the number of times of blinking of the figure different according to the classification result. The medical image processing device 50 may make a type of the figure different according to the classification result. The type of the figure may be a rectangular frame (bounding box) surrounding the lesion, a circular frame surrounding the lesion, a plurality of parentheses surrounding the lesion, an arrow indicating the lesion, or the like. In addition, in a case in which the medical image processing device 50 outputs a voice indicating the position of the lesion to the voice output device, the medical image processing device 50 makes at least one of a volume, a pitch, the presence or absence of repetition, a cycle of repetition, or the number of times of repetition of the voice different according to the classification result.

[0073] In addition, the medical image processing device 50 may be configured not to provide a notification only in a case of specific class classification. For example, in a case of lesion detection, only a lesion with a relatively low malignancy grade is left unnotified, whereby it is possible to reduce the adverse effect of information with a relatively low degree of importance interfering with a doctor's diagnosis.

Third Embodiment

[0074] An application example of the medical image processing device 50 in which both high accuracy and real-time processing are achieved for the detection and classification of the lesion will be described. FIG. 6 is a schematic diagram showing an overall configuration of an endoscope

system **100** including the medical image processing device **50**. As shown in FIG. 6, the endoscope system **100** comprises the medical image processing device **50**, an endoscope **110** that is an electronic endoscope, a light source device **111**, an endoscope processor device **112**, and a display device **113**.

[0075] The endoscope **110** is used for capturing a medical image which is a time-series image, and is, for example, a flexible endoscope. The endoscope **110** includes an insertion part **120** that is inserted into a subject and that has a distal end and a base end, a hand operation part **121** that is consecutively installed on a base end side of the insertion part **120** and that is held by a doctor for performing various operations, and a universal cord **122** that is consecutively installed with the hand operation part **121**.

[0076] The insertion part **120** is formed in an elongated shape with a small diameter as a whole. The insertion part **120** is configured by consecutively installing, in order from the base end side to a distal end side, a flexible portion **125** that has flexibility, a bendable portion **126** that can be bent by operating the hand operation part **121**, and a distal end portion **127** that incorporates an imaging optical system (not shown), an imaging element **128**, and the like.

[0077] The imaging element **128** is a complementary metal oxide semiconductor (CMOS) type imaging element or a charge coupled device (CCD) type imaging element. Image light of an observed part is incident on an imaging surface of the imaging element **128** via an observation window (not shown) that is open to a distal end surface of the distal end portion **127**, and an objective lens (not shown) disposed behind the observation window. The imaging element **128** images the image light of the observed part incident on its imaging surface and outputs an imaging signal.

[0078] Two types of bending operation knobs **129** that are used for the bending operation of the bendable portion **126**, an air/water supply button **130** for an air/water supply operation, and a suction button **131** for a suction operation are disposed in the hand operation part **121**. In addition, the hand operation part **121** is provided with a still image capturing instruction unit **132** for giving an imaging instruction of a still image **139** of the observed part and a treatment tool inlet port **133** through which a treatment tool (not shown) is inserted into a treatment tool insertion passage (not shown) passing through the insertion part **120**.

[0079] The universal cord **122** is a connection cord for connecting the endoscope **110** to the light source device **111**. The universal cord **122** encompasses a light guide **135**, a signal cable **136**, and a fluid tube (not shown) which are inserted into the insertion part **120**. In addition, a connector **137a** that is connected to the light source device **111**, and a connector **137b** that branches from the connector **137a** and that is connected to the endoscope processor device **112** are disposed in an end part of the universal cord **122**.

[0080] By connecting the connector **137a** to the light source device **111**, the light guide **135** and the fluid tube are inserted into the light source device **111**. Accordingly, necessary illumination light, water, and gas are supplied from the light source device **111** to the endoscope **110** via the light guide **135** and the fluid tube. As a result, illumination light is emitted from an illumination window (not shown) on the distal end surface of the distal end portion **127** toward the observed part. In addition, a gas or water is jetted from an air/water supply nozzle (not shown) on the distal end surface of the distal end portion **127** toward an observation window (not shown) on the distal end surface according to a pressing operation of the air/water supply button **130** described above.

[0081] By connecting the connector **137b** to the endoscope processor device **112**, the signal cable **136** and the endoscope processor device **112** are electrically connected to each other. Accordingly, via the signal cable **136**, the imaging signal of the observed part is output from the imaging element **128** of the endoscope **110** to the endoscope processor device **112**, and a control signal is output from the endoscope processor device **112** to the endoscope **110**.

[0082] The light source device **111** supplies the illumination light to the light guide **135** of the endoscope **110** via the connector **137a**. As the illumination light, white light which is light in a

white wavelength range or light in a plurality of wavelength ranges, light in one or a plurality of specific wavelength ranges, or light in various wavelength ranges according to an observation purpose, such as a combination of these, is selected. Note that the specific wavelength range is a range narrower than the white wavelength range.

[0083] A first example of the specific wavelength range is, for example, a blue or green range of visible range. The wavelength range of the first example includes a wavelength range of 390 nm or more and 450 nm or less or 530 nm or more and 550 nm or less, and light of the first example has a peak wavelength in the wavelength range of 390 nm or more and 450 nm or less or 530 nm or more and 550 nm or less.

[0084] A second example of the specific wavelength range is, for example, a red range of the visible range. The wavelength range of the second example includes a wavelength range of 585 nm or more and 615 nm or less or 610 nm or more and 730 nm or less, and light of the second example has a peak wavelength in the wavelength range of 585 nm or more and 615 nm or less or 610 nm or more and 730 nm or less.

[0085] A third example of the specific wavelength range includes a wavelength range of which a light absorption coefficient is different between oxygenated hemoglobin and reduced hemoglobin, and light of the third example has a peak wavelength in the wavelength range of which the light absorption coefficient is different between the oxygenated hemoglobin and the reduced hemoglobin. The wavelength range of the third example includes a wavelength range of 400 ± 10 nm, 440 ± 10 nm, 470 ± 10 nm, or 600 nm or more and 750 nm or less, and light of the third example has a peak wavelength in the wavelength range of 400 ± 10 nm, 440 ± 10 nm, 470 ± 10 nm, or 600 nm or more and 750 nm or less.

[0086] A fourth example of the specific wavelength range is a wavelength range (390 nm to 470 nm) of excitation light that is used for observing (fluorescence observation) fluorescence emitted by a fluorescent substance in a living body and that excites the fluorescent substance.

[0087] A fifth example of the specific wavelength range is a wavelength range of infrared light. The wavelength range of the fifth example includes a wavelength range of 790 nm or more and 820 nm or less or 905 nm or more and 970 nm or less, and light of the fifth example has a peak wavelength in the wavelength range of 790 nm or more and 820 nm or less or 905 nm or more and 970 nm or less.

[0088] The endoscope processor device **112** controls an operation of the endoscope **110** via the connector **137b** and the signal cable **136**. In addition, the endoscope processor device **112** generates a video **138**, which is a time-series medical image consisting of time-series frame images including a subject image, based on the imaging signal acquired from the imaging element **128** of the endoscope **110** via the connector **137b** and the signal cable **136**. A frame rate of the video **138** is, for example, 30 frames per second (fps).

[0089] Further, in a case in which the still image capturing instruction unit **132** is operated by the hand operation part **121** of the endoscope **110**, the endoscope processor device **112** acquires one frame image of the video **138** in accordance with a timing of imaging instruction, in parallel with the generation of the video **138**, and sets the frame image as the still image **139**.

[0090] The endoscope processor device **112** outputs the generated video **138** and the generated still image **139** to the display device **113** and the medical image processing device **50** in real time.

[0091] The endoscope processor device **112** may generate (acquire) a special light image having information on the specific wavelength range based on a normal light image obtained by the white light. The endoscope processor device **112** obtains a signal of the specific wavelength range by performing calculation based on RGB color information of red, green, and blue or CMY color information of cyan, magenta, and yellow, which is contained in the normal light image.

[0092] In addition, the endoscope processor device **112** may generate a feature amount image such as a known oxygen saturation image, for example, based on at least one of the normal light image obtained by the white light or the special light image obtained by the light (special light) of the

specific wavelength range. In this case, the endoscope processor device **112** functions as a feature amount image generation unit. The video **138** or the still image **139** including the image of the inside of the living body, the normal light image, the special light image, and the feature amount image are medical images obtained by visualizing a result of imaging or measuring a human body for diagnosis and testing purposes based on images.

[0093] The display device **113** is connected to the endoscope processor device **112** and displays the video **138** and the still image **139** input from the endoscope processor device **112**. The doctor performs a forward and backward operation or the like of the insertion part **120** while checking the video **138** displayed on the display device **113**. In a case in which a lesion or the like is found in the observed part, the doctor executes capturing of a still image of the observed part by operating the still image capturing instruction unit **132** and performs a diagnosis, a biopsy, and the like.

[0094] With the endoscope system **100** to which the medical image processing device **50** is applied, it is possible to detect and classify a lesion in real time with high accuracy from the video **138** and the still image **139** captured by the imaging element **128** of the endoscope **110**.

Others

[0095] Although the hierarchical neural network that performs the detection and the class classification of the lesion from the medical image has been described here, the present disclosure is not limited to the medical image and can be applied to various images. For example, the technique according to the present disclosure can be applied to a case in which a region of interest is detected from an image in which a structure such as a bridge is imaged, and the region of interest is classified into fissuring, peeling, rust, a hole, and the like.

[0096] The technical scope of the present invention is not limited to the scope described in the above embodiments. The configurations and the like in each embodiment can be appropriately combined among the respective embodiments without departing from the spirit of the present invention.

Explanation of References

[0097] **10**: network system [0098] **12**: feature extraction unit [0099] **14**: detection processing unit [0100] **20**: network system [0101] **22**: feature extraction unit [0102] **24**: detection processing unit [0103] **26**: cropping unit [0104] **28**: classification processing unit [0105] **30**: network system [0106] **36**: cropping unit [0107] **38**: resizing unit [0108] **40**: classification processing unit [0109] **50**: medical image processing device [0110] **52**: processor [0111] **54**: memory [0112] **56**: input device [0113] **58**: output device [0114] **100**: endoscope system [0115] **110**: endoscope [0116] **111**: light source device [0117] **112**: endoscope processor device [0118] **113**: display device [0119] **120**: insertion part [0120] **121**: hand operation part [0121] **122**: universal cord [0122] **125**: flexible portion [0123] **126**: bendable portion [0124] **127**: distal end portion [0125] **128**: imaging element [0126] **129**: bending operation knob [0127] **130**: air/water supply button [0128] **131**: suction button [0129] **132**: still image capturing instruction unit [0130] **133**: treatment tool inlet port [0131] **135**: light guide [0132] **136**: signal cable [0133] **137a**: connector [0134] **137b**: connector [0135] **138**: video [0136] **139**: still image [0137] **BB**: bounding box [0138] **FM1**, **FM2**, **FM3**: feature map [0139] **FV1**, **FV2**, **FV3**, **FV4**: feature amount [0140] **FV11**, **FV12**, **FV13**, **FV14**: feature amount [0141] **FV21**, **FV22**, **FV23**: feature amount [0142] **S1** to **S6**: step of medical image processing method

Claims

1. A medical image processing device comprising: one or more processors that acquire a medical image; and one or more memories that store a program to be executed by the one or more processors, wherein the one or more processors extract a first feature amount and a second feature amount having a resolution relatively higher than a resolution of the first feature amount from the medical image by processing the medical image in a feature extraction network of a hierarchical

neural network including the feature extraction network, a first subnetwork, and a second subnetwork, detect a region of interest included in the medical image by processing the first feature amount in the first subnetwork of the hierarchical neural network, and classify the region of interest by processing the second feature amount in the second subnetwork of the hierarchical neural network.

2. The medical image processing device according to claim 1, wherein the second feature amount is an intermediate feature amount in a process of extracting the first feature amount from the medical image in the feature extraction network.

3. The medical image processing device according to claim 1, wherein the one or more processors train the first subnetwork using a first data set and train the second subnetwork using a second data set, the first data set includes a set of a first medical image and position information of a region of interest included in the first medical image, and the second data set includes a set of a second medical image and position information and a classification class label of a region of interest included in the second medical image.

4. The medical image processing device according to claim 3, wherein the one or more processors train the feature extraction network and the first subnetwork using the first data set, and train the second subnetwork using the second data set based on the trained feature extraction network and the trained first subnetwork.

5. The medical image processing device according to claim 4, wherein the one or more processors pre-train the feature extraction network using a third data set different from the first data set and the second data set before training the feature extraction network and the first subnetwork using the first data set.

6. The medical image processing device according to claim 1, wherein the one or more processors notify of position information of the region of interest in a manner corresponding to a result of the classification.

7. The medical image processing device according to claim 6, wherein the one or more processors add information based on the position information to the medical image and display the medical image on a display.

8. The medical image processing device according to claim 6, wherein the one or more processors do not notify of the position information of the region of interest in a case in which the result of the classification is a specific class.

9. The medical image processing device according to claim 8, wherein classes of the classification include a malignancy grade, and the one or more processors do not notify of the position information of the region of interest in a case in which the malignancy grade is relatively low.

10. The medical image processing device according to claim 1, wherein the one or more processors crop a part of the second feature amount according to a detection result of the region of interest, and process the cropped second feature amount in the second subnetwork.

11. The medical image processing device according to claim 10, wherein the one or more processors align a size of the cropped second feature amount in a spatial direction to a certain size, and process the second feature amount having the certain size in the second subnetwork.

12. A hierarchical neural network comprising: a feature extraction network that extracts a first feature amount and a second feature amount having a resolution relatively higher than a resolution of the first feature amount from an input medical image; a first subnetwork that detects a region of interest included in the medical image from the input first feature amount; and a second subnetwork that classifies the region of interest from the input second feature amount.

13. A medical image processing method executed by one or more processors, the medical image processing method comprising: acquiring a medical image; extracting a first feature amount and a second feature amount having a resolution relatively higher than a resolution of the first feature amount from the medical image by processing the medical image in a feature extraction network of a hierarchical neural network including the feature extraction network, a first subnetwork, and a

second subnetwork; detecting a region of interest included in the medical image by processing the first feature amount in the first subnetwork of the hierarchical neural network; and classifying the region of interest by processing the second feature amount in the second subnetwork of the hierarchical neural network.

14. A non-transitory, computer-readable tangible recording medium which records thereon, a program for causing, when read by a computer, one or more processors of the computer to execute the medical image processing method according to claim 13.
